UCI ADULT INCOME
BASELINES, ML MODEL & INITIAL ERROR ANALYSIS

PROBLEM FRAMING
Predict whether an adult earns more than $50K per year.
The business scenario is targeted higher-value outreach.
Precision is the primary metric because false positives
represent wasted outreach resources.

DATASET
Total rows: 48,842
Positive class (>50K): 23.93%

TRAIN / DEV / TEST
Train: 34,188 (70.0%)
Dev: 4,885 (10.0%)
Test: 9,769 (20.0%)

Split is stratified with random_state=42.
The final test set was kept separate from model development.

BASELINE RESULTS

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority class	0.761	0.000	0.000	0.000	0.500	0.239
Education >= 13	0.753	0.484	0.497	0.491	0.665	0.361

FINAL ML MODEL

HistGradientBoosting

Accuracy: 0.876
Precision: 0.794
Recall: 0.650
F1: 0.715
ROC AUC: 0.928
PR AUC: 0.830

FINAL TEST ACCURACY: 87.59%

ERROR ANALYSIS

False positives: 393
False negatives: 819

The ML model can make mistakes because income depends on multiple
factors rather than a single feature. Important patterns include
education, age, occupation, work hours, marital status and capital
gains/losses.

NEXT STEPS

- Improve categorical missing-value handling.
- Experiment with encoding strategies.
- Handle skewed capital-gain and capital-loss.
- Explore nonlinear feature interactions.
- Tune the model using the development set.
- Monitor precision, recall, F1 and PR AUC.
- Evaluate subgroup/fairness performance.

PRIMARY SUCCESS METRIC

PRECISION

The final model should improve positive-class precision while
maintaining useful recall and PR AUC. Accuracy above 85% is a useful
technical target, but it should not replace the primary business
objective of reducing wasted positive predictions.